In [1]:
```
# VAR robustness (Section 4.8): Table 6, Tables B.2, C.3, C.4 and C.7, Section 5.1
import os
import warnings
import numpy as np
import pandas as pd
from statsmodels.tsa.vector_ar.var_model import VAR

warnings.filterwarnings("ignore")
os.makedirs("output", exist_ok=True)
```

In [2]:
```
# Merge the clean files into one monthly data set, December 2015 to December 2025 (121
months)
files = ["spy_clean.csv", "rsp_clean.csv", "voo_clean.csv", "vix_clean.csv",
"dgs10_clean.csv",
         "AIStocks_ASVI8_Monthly.csv", "AIStocks_ASVI4_Monthly.csv",
         "mag7_clean.csv", "mag7b_clean.csv", "mag7c_clean.csv"]
df = pd.read_csv(files[0], parse_dates=["date"])
for f in files[1:]:
    df = df.merge(pd.read_csv(f, parse_dates=["date"]), on="date", how="inner")
df = df.sort_values("date").reset_index(drop=True)

# 2021-2025 subsample. December 2020 only serves as the first lag, so the VARs run on Jan
2021 to Dec 2025.
sub = df[df["date"] >= "2020-12-31"].reset_index(drop=True)
print(len(df), "months,", df["date"].min().date(), "to", df["date"].max().date())

labels = {"dlog_vix": "dlogVIX", "d_dgs10": "dDGS10", "ASVI8": "ASVI8", "ASVI4": "ASVI4",
          "flow_spy": "Flow SPY", "flow_rsp": "Flow RSP", "flow_voo": "Flow VOO",
          "mag7_ret": "Mag7 returns", "mag7b_ret": "Mag7B returns", "mag7c_ret": "Mag7C
returns"}

samples = {"Full sample": df, "2021-2025": sub}
```

```
121 months, 2015-12-31 to 2025-12-31
```

In [3]:
```
def stars(p):
    return "***" if p < 0.01 else "**" if p < 0.05 else "*" if p < 0.10 else ""

def granger(res, cause, effect):
    # F-test: do the lags of `cause` help predict `effect`?
    t = res.test_causality(effect, [cause], kind="f")
    return float(t.test_statistic), float(t.pvalue)


def fevd(res, response):
    # Share (%) of the forecast error variance of `response` explained by each shock, h = 1,
6, 12
    d = res.fevd(12).decomp[list(res.names).index(response)] * 100
    return pd.DataFrame({"h=1": d[0], "h=6": d[5], "h=12": d[11]}, index=list(res.names))


def irf_bands(res, impulse, response, h=12):
    # Orthogonalized impulse response with asymptotic 95% bands
    i, j = list(res.names).index(response), list(res.names).index(impulse)
    irf = res.irf(h)
    point = irf.orth_irfs[:, i, j]
    se = irf.stderr(orth=True)[:, i, j]
    return point, point - 1.959964 * se, point + 1.959964 * se


def largest_eigenvalue(res):
    # statsmodels reports the roots of the characteristic polynomial. The eigenvalues of the
    # companion matrix are their inverses, so the largest eigenvalue is 1 / smallest root.
    return 1 / np.abs(res.roots).min()
```

In [4]:
```
# 22 specifications, each estimated on both samples:
#   A: Mag7 with the SPY, RSP or VOO flow (4.8.1)
#   B: Mag7B or Mag7C with each flow (4.8.2)
#   C: three-variable VAR without the yield and the flow (4.8.3)
#   D: ASVI4 instead of ASVI8 (4.8.4)
specs = []
```

```
for ret, basket in [("mag7_ret", "Mag7"), ("mag7b_ret", "Mag7B"), ("mag7c_ret", "Mag7C")]:
    for flow in ["flow_spy", "flow_rsp", "flow_voo"]:
        specs.append(("A" if basket == "Mag7" else "B", basket, flow[5:].upper(),
                      ["dlog_vix", "d_dgs10", "ASVI8", flow, ret]))
specs.append(("C", "Mag7, 3-variable", "None", ["dlog_vix", "ASVI8", "mag7_ret"]))
specs.append(("D", "Mag7, ASVI4", "SPY", ["dlog_vix", "d_dgs10", "ASVI4", "flow_spy",
"mag7_ret"]))

results = {}
rows = []
for name, data in samples.items():
    for group, basket, flow, c in specs:
        r = VAR(data[c]).fit(1)
        results[(basket, flow, name)] = r
        att = c[2] if len(c) == 5 else c[1]            # attention variable
        fl = c[3] if len(c) == 5 else None             # flow (none in the three-variable
VAR)
        ret = c[-1]                                    # return variable
        f, p = granger(r, att, ret)
        w, n = r.test_whiteness(nlags=10), r.test_normality()
        rows.append({
            "group": group, "basket": basket, "flow": flow, "sample": name, "N": int(r.nobs),
            "beta": r.params.loc["L1." + att, ret], "se": r.stderr.loc["L1." + att, ret],
            "beta_p": r.pvalues.loc["L1." + att, ret], "F": f, "p": p,
            "p_att_to_flow": granger(r, att, fl)[1] if fl else np.nan,
            "p_flow_to_ret": granger(r, fl, ret)[1] if fl else np.nan,
            "p_ret_to_att": granger(r, ret, att)[1],
            "fevd_att": fevd(r, ret).loc[att, "h=12"],
            "fevd_flow": fevd(r, ret).loc[fl, "h=12"] if fl else np.nan,
            "bic_lag": int(VAR(data[c]).select_order(maxlags=4).selected_orders["bic"]),
            "max_eig": largest_eigenvalue(r), "Q": w.test_statistic, "Q_p": w.pvalue,
            "normality": n.test_statistic, "normality_p": n.pvalue,
            "own_lag_att": r.params.loc["L1." + att, att],
        })
specs = pd.DataFrame(rows)
asvi8 = specs[specs["basket"] != "Mag7, ASVI4"]
print(len(specs), "VARs estimated")
```

```
22 VARs estimated
```

In [5]:

```
# Table 6
def spec_label(row):
    if row.group == "A":
        return f"{row.flow}, {row['sample']}"
    if row.group == "B":
        return f"{row.basket} ({'excl. TSLA' if row.basket == 'Mag7B' else 'AI-pure'}),
{row['sample']}"
    if row.group == "C":
        return f"3-var, {row['sample']}"
    return f"ASVI4, SPY, {row['sample']}"

t6 = specs[specs["group"].isin(["A", "C", "D"]) | ((specs["group"] == "B") & (specs["flow"]
== "SPY"))].copy()
t6["s"] = t6["sample"].map({"Full sample": 0, "2021-2025": 1})
t6["b"] = t6["basket"].map({"Mag7": 0, "Mag7B": 1, "Mag7C": 2}).fillna(0)
t6["f"] = t6["flow"].map({"SPY": 0, "RSP": 1, "VOO": 2, "None": 0})
t6 = t6.sort_values(["group", "s", "b", "f"])          # row order of Table 6

table6 = pd.DataFrame({
    "Panel": t6["group"].values,
    "Specification": [spec_label(r) for _, r in t6.iterrows()],
    "beta": [f"{x:.4f}" for x in t6["beta"]],
    "SE": [f"({x:.4f})" for x in t6["se"]],
    "Granger F": [f"{f:.3f}{stars(p)}" for f, p in zip(t6["F"], t6["p"])],
    "p": [f"{x:.3f}" for x in t6["p"]],
    "FEVD h=12": [f"{x:.1f}%" for x in t6["fevd_att"]],
    "N": t6["N"].values})
table6.to_csv("output/Table_6_robustness_summary.csv", index=False)
print("ASVI8 specifications significant at 5%:", (asvi8["p"] < 0.05).sum(), "of", len(asvi8))
table6
```

```
ASVI8 specifications significant at 5%: 20 of 20
```

Out[5]:

	Panel	Specification	beta	SE	Granger F	p	FEVD h=12	N
0	A	SPY, Full sample	0.0274	(0.0128)	4.625**	0.032	4.8%	120
1	A	RSP, Full sample	0.0269	(0.0127)	4.491**	0.035	4.8%	120
2	A	VOO, Full sample	0.0274	(0.0127)	4.659**	0.031	4.8%	120
3	A	SPY, 2021-2025	0.0378	(0.0161)	5.500**	0.020	10.9%	60
4	A	RSP, 2021-2025	0.0385	(0.0160)	5.833**	0.016	11.1%	60
5	A	VOO, 2021-2025	0.0397	(0.0157)	6.384**	0.012	10.7%	60
6	B	Mag7B (excl. TSLA), Full sample	0.0306	(0.0118)	6.682***	0.010	6.1%	120
7	B	Mag7C (AI-pure), Full sample	0.0349	(0.0127)	7.549***	0.006	8.0%	120
8	B	Mag7B (excl. TSLA), 2021-2025	0.0394	(0.0153)	6.611**	0.011	13.3%	60
9	B	Mag7C (AI-pure), 2021-2025	0.0413	(0.0172)	5.723**	0.017	14.6%	60
10	C	3-var, Full sample	0.0275	(0.0126)	4.778**	0.029	5.2%	120
11	C	3-var, 2021-2025	0.0386	(0.0156)	6.092**	0.015	10.8%	60
12	D	ASVI4, SPY, Full sample	0.0164	(0.0149)	1.204	0.273	3.3%	120
13	D	ASVI4, SPY, 2021-2025	0.0225	(0.0193)	1.354	0.246	6.4%	60

In [6]:

```
# Table B.2: diagnostics of all specifications
tableB2 = pd.DataFrame({
    "Sample": specs["sample"], "Specification": specs["basket"], "Flow": specs["flow"],
    "BIC lag": specs["bic_lag"],
    "Max. eigenvalue modulus": [f"{x:.3f}" for x in specs["max_eig"]],
    "Portmanteau Q": [f"{x:.1f}" for x in specs["Q"]],
    "p (Q)": [f"{x:.3f}" for x in specs["Q_p"]],
    "Normality": [f"{x:.2f}" for x in specs["normality"]],
    "p (normality)": ["< 0.001" if x < 0.001 else f"{x:.3f}" for x in specs["normality_p"]]})
tableB2.to_csv("output/Table_B2_diagnostics_all_specifications.csv", index=False)
print("ASVI8 specifications: stable", (asvi8["max_eig"] < 1).sum(), "| no residual
autocorrelation at 5%",
      (asvi8["Q_p"] > 0.05).sum(), "| BIC lag 1", (asvi8["bic_lag"] == 1).sum(),
      "| normality not rejected", (asvi8["normality_p"] > 0.05).sum(), "(of 20)")
tableB2
```

```
ASVI8 specifications: stable 20 | no residual autocorrelation at 5% 19 | BIC lag 1 20 |
normality not rejected 4 (of 20)
```

Out[6]:

	Sample	Specification	Flow	BIC lag	Max. eigenvalue modulus	Portmanteau Q	p (Q)	Normality	p (normality)
0	Full sample	Mag7	SPY	1	0.793	234.9	0.312	57.65	< 0.001
1	Full sample	Mag7	RSP	1	0.789	241.0	0.221	110.24	< 0.001
2	Full sample	Mag7	VOO	1	0.791	240.1	0.233	29.02	0.001
3	Full sample	Mag7B	SPY	1	0.794	229.2	0.410	59.46	< 0.001
4	Full sample	Mag7B	RSP	1	0.791	231.0	0.378	112.55	< 0.001
5	Full sample	Mag7B	VOO	1	0.792	238.2	0.260	29.18	0.001
6	Full sample	Mag7C	SPY	1	0.793	228.2	0.428	59.44	< 0.001
7	Full sample	Mag7C	RSP	1	0.790	236.9	0.280	112.28	< 0.001
8	Full sample	Mag7C	VOO	1	0.791	241.0	0.221	28.55	0.001
9	Full sample	Mag7, 3-variable	None	1	0.793	108.5	0.023	27.81	< 0.001
10	Full sample	Mag7, ASVI4	SPY	0	0.636	248.9	0.131	47.92	< 0.001
11	2021-2025	Mag7	SPY	1	0.838	221.0	0.562	51.30	< 0.001
12	2021-2025	Mag7	RSP	1	0.828	211.6	0.730	28.47	0.002
13	2021-2025	Mag7	VOO	1	0.828	233.7	0.331	14.33	0.158
14	2021-2025	Mag7B	SPY	1	0.838	219.9	0.584	51.31	< 0.001
15	2021-2025	Mag7B	RSP	1	0.828	213.1	0.706	27.64	0.002
16	2021-2025	Mag7B	VOO	1	0.828	241.0	0.221	12.78	0.236
17	2021-2025	Mag7C	SPY	1	0.836	228.3	0.427	51.18	< 0.001
18	2021-2025	Mag7C	RSP	1	0.828	225.3	0.482	29.58	0.001
19	2021-2025	Mag7C	VOO	1	0.826	247.6	0.143	12.54	0.250
20	2021-2025	Mag7, 3-variable	None	1	0.838	95.4	0.131	7.93	0.244
21	2021-2025	Mag7, ASVI4	SPY	0	0.679	221.7	0.550	48.57	< 0.001

In [7]:

```
# Table C.3: Granger tests and variance decomposition of all specifications
def blank_if_missing(x, d):
    return "-" if pd.isna(x) else f"{x:.{d}f}"

tableC3 = pd.DataFrame({
    "Sample": specs["sample"], "Specification": specs["basket"], "Flow": specs["flow"],
    "beta on L1.ASVI (SE)": [f"{b:.4f}{stars(p)} ({s:.4f})" for b, s, p in zip(specs["beta"],
specs["se"], specs["beta_p"])],
    "F": [f"{x:.3f}" for x in specs["F"]],
    "p": [f"{x:.3f}" for x in specs["p"]],
    "p: ASVI to flow": [blank_if_missing(x, 3) for x in specs["p_att_to_flow"]],
    "p: flow to return": [blank_if_missing(x, 3) for x in specs["p_flow_to_ret"]],
    "p: return to ASVI": [f"{x:.3f}" for x in specs["p_ret_to_att"]],
    "FEVD ASVI (%)": [f"{x:.1f}" for x in specs["fevd_att"]],
    "FEVD flow (%)": [blank_if_missing(x, 1) for x in specs["fevd_flow"]]})
tableC3.to_csv("output/Table_C3_granger_fevd_all_specifications.csv", index=False)

# Section 5.1: the 18 specifications with a flow variable and ASVI8
with_flow = asvi8[asvi8["flow"] != "None"]
```

```
print(f"ASVI8 -> flow p-values: {with_flow['p_att_to_flow'].min():.3f} to
{with_flow['p_att_to_flow'].max():.3f}, "
      f"flow -> return significant at 5% in {(with_flow['p_flow_to_ret'] < 0.05).sum()} of
{len(with_flow)}")
tableC3
```

```
ASVI8 -> flow p-values: 0.200 to 0.994, flow -> return significant at 5% in 0 of 18
```

Out[7]:

	Sample	Specification	Flow	beta on L1.ASVI (SE)	F	p	p: ASVI to flow	p: flow to return	p: return to ASVI	FEVD ASVI (%)	FEVD flow (%)
0	Full sample	Mag7	SPY	0.0274** (0.0128)	4.625	0.032	0.210	0.976	0.192	4.8	0.5
1	Full sample	Mag7	RSP	0.0269** (0.0127)	4.491	0.035	0.913	0.483	0.161	4.8	0.5
2	Full sample	Mag7	VOO	0.0274** (0.0127)	4.659	0.031	0.925	0.765	0.163	4.8	1.2
3	Full sample	Mag7B	SPY	0.0306*** (0.0118)	6.682	0.010	0.219	0.984	0.202	6.1	0.3
4	Full sample	Mag7B	RSP	0.0304*** (0.0118)	6.656	0.010	0.959	0.720	0.185	6.2	0.5
5	Full sample	Mag7B	VOO	0.0305*** (0.0118)	6.732	0.010	0.884	0.611	0.154	5.9	2.3
6	Full sample	Mag7C	SPY	0.0349*** (0.0127)	7.549	0.006	0.200	0.942	0.099	8.0	0.3
7	Full sample	Mag7C	RSP	0.0348*** (0.0127)	7.558	0.006	0.936	0.984	0.089	8.2	0.5
8	Full sample	Mag7C	VOO	0.0347*** (0.0126)	7.558	0.006	0.774	0.558	0.072	7.7	2.2
9	Full sample	Mag7, 3-variable	None	0.0275** (0.0126)	4.778	0.029	-	-	0.144	5.2	-
10	Full sample	Mag7, ASVI4	SPY	0.0164 (0.0149)	1.204	0.273	0.286	0.880	0.376	3.3	0.3
11	2021-2025	Mag7	SPY	0.0378** (0.0161)	5.500	0.020	0.379	0.693	0.511	10.9	0.2
12	2021-2025	Mag7	RSP	0.0385** (0.0160)	5.833	0.016	0.900	0.783	0.452	11.1	0.5
13	2021-2025	Mag7	VOO	0.0397** (0.0157)	6.384	0.012	0.643	0.166	0.427	10.7	3.5
14	2021-2025	Mag7B	SPY	0.0394** (0.0153)	6.611	0.011	0.411	0.920	0.389	13.3	0.0
15	2021-2025	Mag7B	RSP	0.0396*** (0.0152)	6.811	0.010	0.994	0.963	0.404	13.7	1.1
16	2021-2025	Mag7B	VOO	0.0404*** (0.0150)	7.275	0.007	0.541	0.232	0.297	12.0	2.9
17	2021-2025	Mag7C	SPY	0.0413** (0.0172)	5.723	0.017	0.465	0.967	0.210	14.6	0.1
18	2021-2025	Mag7C	RSP	0.0411** (0.0171)	5.778	0.017	0.900	0.806	0.236	15.4	1.8
19	2021-2025	Mag7C	VOO	0.0418** (0.0169)	6.103	0.014	0.495	0.279	0.162	13.2	2.3
20	2021-2025	Mag7, 3-variable	None	0.0386** (0.0156)	6.092	0.015	-	-	0.464	10.8	-
21	2021-2025	Mag7, ASVI4	SPY	0.0225 (0.0193)	1.354	0.246	0.235	0.555	0.463	6.4	0.7

In [8]:

```
# Table C.4: variance decomposition of Mag7 returns in VAR-RSP and VAR-VOO (%)
tableC4 = pd.concat({
    f"VAR-{flow}": pd.concat({name: fevd(results[("Mag7", flow, name)],
"mag7_ret").rename(index=labels)
                              for name in samples}, axis=1)
    for flow in ["RSP", "VOO"]}).round(1)
tableC4.to_csv("output/Table_C4_FEVD_RSP_VOO.csv")
tableC4
```

Out[8]:

		Full sample			2021-2025		
		h=1	h=6	h=12	h=1	h=6	h=12
VAR-RSP	**dlogVIX**	28.0	27.1	27.0	24.9	23.3	23.1
	dDGS10	2.2	2.3	2.3	7.5	6.9	6.8
	ASVI8	1.9	4.6	4.8	3.3	10.1	11.1
	Flow RSP	0.0	0.5	0.5	0.0	0.4	0.5
	Mag7 returns	67.8	65.5	65.3	64.4	59.3	58.5
VAR-VOO	**dlogVIX**	28.2	27.3	27.3	22.3	21.7	21.7
	dDGS10	2.3	2.4	2.4	7.7	7.0	6.9
	ASVI8	1.9	4.5	4.8	4.8	10.0	10.7
	Flow VOO	1.0	1.2	1.2	0.8	3.4	3.5
	Mag7 returns	66.5	64.6	64.4	64.3	57.9	57.1

In [9]:

```
# Numbers in Sections 4.8.2 to 4.8.4
spy = specs[(specs["flow"] == "SPY") & specs["basket"].isin(["Mag7", "Mag7B", "Mag7C"])]
print(spy.pivot(index="basket", columns="sample", values=["beta", "F", "fevd_att"]).round(3))

mag7c = specs[(specs["basket"] == "Mag7C") & (specs["sample"] == "Full sample")]
print("\nMag7C, full sample, return -> ASVI8 p:", dict(zip(mag7c["flow"],
mag7c["p_ret_to_att"].round(3))))

three = specs[specs["basket"] == "Mag7, 3-variable"].set_index("sample")
five = specs[(specs["basket"] == "Mag7") & (specs["flow"] == "SPY")].set_index("sample")
print("\n3-variable VAR: beta", three["beta"].round(4).to_dict(), "| whiteness p",
three["Q_p"].round(3).to_dict())
print("Largest difference to the 5-variable beta:", round((three["beta"] -
five["beta"]).abs().max(), 4))

a4 = specs[specs["basket"] == "Mag7, ASVI4"].set_index("sample")
print("\nOwn lag, full sample: ASVI4", round(a4.loc["Full sample", "own_lag_att"], 3),
      "| ASVI8", round(five.loc["Full sample", "own_lag_att"], 3))
```

```
           beta                      F             fevd_att
sample 2021-2025 Full sample 2021-2025 Full sample 2021-2025 Full sample
basket
Mag7       0.038       0.027     5.500       4.625    10.937       4.830
Mag7B      0.039       0.031     6.611       6.682    13.304       6.108
Mag7C      0.041       0.035     5.723       7.549    14.566       7.952

Mag7C, full sample, return -> ASVI8 p: {'SPY': 0.099, 'RSP': 0.089, 'VOO': 0.072}

3-variable VAR: beta {'Full sample': 0.0275, '2021-2025': 0.0386} | whiteness p {'Full sample':
0.023, '2021-2025': 0.131}
Largest difference to the 5-variable beta: 0.0008

Own lag, full sample: ASVI4 0.627 | ASVI8 0.771
```

In [10]:

```
# Table C.7: Cholesky ordering. The alternative puts ASVI8 last, so attention cannot move
# any other variable within the same month.
orderings = {"Baseline": ["dlog_vix", "d_dgs10", "ASVI8", "flow_spy", "mag7_ret"],
             "ASVI8 last": ["dlog_vix", "d_dgs10", "flow_spy", "mag7_ret", "ASVI8"]}
rows = []
for name, data in samples.items():
    for label, c in orderings.items():
        r = VAR(data[c]).fit(1)
        fe = fevd(r, "mag7_ret").loc["ASVI8"]
        point, lower, upper = irf_bands(r, "ASVI8", "mag7_ret")
        rows.append({"Sample": name, "Ordering": label,
                     "FEVD h=1 (%)": f"{fe['h=1']:.1f}", "FEVD h=6 (%)": f"{fe['h=6']:.1f}",
                     "FEVD h=12 (%)": f"{fe['h=12']:.1f}", "IRF h=0 (%)": f"{point[0] *
100:.3f}",
                     "IRF h=1 (%) [95% CI]": f"{point[1] * 100:.3f} [{lower[1] * 100:.3f},
{upper[1] * 100:.3f}]"})
tableC7 = pd.DataFrame(rows)
tableC7.to_csv("output/Table_C7_Cholesky_ordering.csv", index=False)
tableC7
```

Out[10]:

	Sample	Ordering	FEVD h=1 (%)	FEVD h=6 (%)	FEVD h=12 (%)	IRF h=0 (%)	IRF h=1 (%) [95% CI]
0	Full sample	Baseline	1.9	4.6	4.8	1.008	0.802 [-0.033, 1.637]
1	Full sample	ASVI8 last	0.0	3.0	3.3	0.000	0.915 [0.073, 1.757]
2	2021-2025	Baseline	2.8	9.8	10.9	1.305	1.374 [0.198, 2.551]
3	2021-2025	ASVI8 last	0.0	6.9	7.9	0.000	1.395 [0.203, 2.587]